

Co-Designing a Multimodal AR Assistant for Interventional Radiology Workflows

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Abstract

Interventional radiology (IR) relies on real-time medical imaging to guide minimally invasive procedures under strict sterile constraints. While prior work has explored augmented reality (AR) for IR, less attention has been paid to how interaction design and workflow structure shape the practical use of head-worn AR during live procedures. We present PRISM, an Apple Vision Pro-based multimodal AR assistant co-designed with interventional radiologists to support sterile, in-procedure imaging interaction. Prism consolidates multi-screen imaging into an in-field XR workspace, supports sterile-first voice and hand-gesture input for common imaging operations, and provides a phase-specific guided view for needle insertion designed to be glanceable and low-occlusion. Through clinician-led co-design and iterative prototype walkthroughs, we derive early design insights into workflow-centered AR interaction, sterile-first multimodal task mapping, and visually restrained XR information layout for high-stakes clinical environments.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; *Interaction techniques*; *User centered design*; Empirical studies in HCI.

Keywords

Augmented Reality, Multimodal Interaction, Sterile Interaction, Workflow-Centered Design, Co-Design, Medical Imaging, Clinical HCI

ACM Reference Format:

Binyan Xu and Wei Wu. 2026. Co-Designing a Multimodal AR Assistant for Interventional Radiology Workflows. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3799005>

1 Introduction

Interventional radiology (IR) relies on continuous interpretation of medical imaging to guide instruments during minimally invasive procedures [7, 8]. Unlike surgical contexts where anatomy is directly visible, IR clinicians must coordinate physical manipulation with constant reference to fluoroscopy, ultrasound, or CT images,

often under strict time pressure and sterile constraints [19, 30, 39]. As a result, interaction with imaging systems is central to procedural work.

In current IR suites, imaging interaction is typically distributed across multiple external displays and control systems [20, 26, 33]. Clinicians frequently shift attention between the procedural field and wall-mounted monitors, request view changes from technologists, or rely on non-sterile input devices that are not directly accessible while scrubbed in [6, 25, 37]. These interaction patterns introduce repeated context switches, coordination overhead, and attentional disruption, particularly during phases that require sustained focus and fine motor control, such as needle targeting and insertion.

Head-worn augmented and extended reality (AR/XR) systems have been proposed as a way to bring imaging into the clinician's field of view. However, deploying AR in live IR procedures raises interaction challenges that extend beyond visualization. Sterile environments constrain the use of conventional input devices; headset field-of-view and occlusion risks limit how much information can be displayed; and unreliable or overly rich interaction can interfere with attention rather than support it. These constraints suggest that the value of AR in IR depends less on visual novelty and more on careful alignment between workflow demands, interaction modalities, and information presentation.

Prior work has demonstrated the technical feasibility of AR systems in interventional radiology [3, 10, 11, 15], but interaction design is often treated as secondary to visualization or tracking accuracy. In particular, there is limited articulation of how multimodal input should be mapped to procedural tasks, how information should appear and disappear across workflow phases, and how AR interfaces can remain visually restrained during high-attention moments. From an HCI perspective, this leaves open questions about how AR can be integrated into IR practice as a usable, in-procedure interaction space rather than an auxiliary display.

In this paper, we present PRISM, a Vision Pro-based multimodal AR assistant co-designed with interventional radiologists to support sterile, in-procedure imaging interaction. Prism consolidates multi-screen imaging into an in-field XR workspace, supports sterile-first voice and hand-gesture input for common imaging operations, and provides a phase-specific guided view for needle insertion designed to be glanceable and low-occlusion. Rather than evaluating clinical efficacy, our contribution lies in articulating workflow-centered design choices and early interaction insights that inform how head-worn AR systems can be designed for high-stakes, sterile clinical environments.



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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3799005>

2 Related Work

2.1 Augmented Reality Systems for Interventional Radiology

Interventional radiology (IR) has been an active testbed for augmented reality (AR) systems that aim to bring imaging and spatial guidance closer to the procedural field. Early and recent systems have explored optical see-through head-mounted displays (HMDs) to visualize two-dimensional (2D) image streams and three-dimensional (3D) anatomical reconstructions during procedures such as needle insertion, ablation, vascular access, and endovascular interventions [1, 2, 5, 23, 28, 38]. These systems typically leverage pre-procedural CT or cone-beam CT data to generate holographic overlays that support spatial reasoning and targeting [16, 32].

Clinical feasibility studies report that AR-assisted workflows can achieve procedural accuracy comparable to conventional image-guided techniques, while in some cases reducing radiation exposure, contrast usage, or procedure time [4, 16, 21]. In addition to live AR overlays, several commercial and research systems provide virtual injection or roadmap-style visualizations that function as navigation adjuncts for IR procedures [4, 13]. Collectively, these studies suggest that AR can meaningfully support image-guided intervention; however, most evaluations emphasize technical feasibility and outcome measures rather than interaction design or workflow integration [15].

2.2 Ergonomics, Attention, and Cognitive Load in AR-Supported Workflows

A major motivation for introducing AR into IR workflows is the high prevalence of work-related musculoskeletal strain and cognitive burden among interventionalists [9, 12]. Conventional IR practice often requires sustained head rotation and repeated gaze shifts between the procedural field and distant imaging monitors, contributing to neck, shoulder, and eye fatigue [9]. By spatially aligning imaging content with the operator's field of view, AR systems have been shown to reduce visual focus shifts and improve ergonomic posture during image-guided tasks [29].

However, ergonomic benefits are not automatic. Prior work highlights practical limitations such as restricted headset field-of-view, visual clutter, occlusion of hands or instruments, headset weight, and system latency, which can negatively impact usability and user comfort if not carefully managed in clinical settings [9, 15]. Studies also report variability in user tolerance and adaptation, including symptoms of eye strain or cybersickness in extended use [17]. These findings underscore that ergonomic improvements depend not only on visualization placement, but also on careful information layout and interaction design that respect attentional limits in high-stakes clinical settings [15].

2.3 Interaction Modalities and Design Gaps in AR-IR Systems

Sterile clinical environments constrain the use of conventional input devices, motivating the adoption of touchless and multimodal interaction techniques for intra-procedure imaging control. Prior AR systems in IR and related surgical domains have explored voice

commands, hand gestures, gaze-based selection, and hybrid combinations for tasks such as view switching, image navigation, and tool alignment [3, 28]. Reviews of multimodal interaction patterns suggest that voice input is commonly used for discrete commands (e.g., mode or series switching), while hand-based input supports continuous navigation and manipulation tasks [15, 28].

Despite these advances, interaction design is often treated as a secondary concern. Many systems rely on preconfigured holographic layouts, external tracking infrastructure, or non-sterile intermediaries, limiting flexibility during live procedures [23, 38]. Moreover, the freedom to place virtual content in three-dimensional space introduces new challenges related to occlusion management, attentional competition, and calibration accuracy [24]. The lack of standardized design frameworks and evaluation metrics further complicates comparison across systems and interaction strategies [15]. These gaps point to the need for workflow-centered, clinician-informed approaches that explicitly address task-to-modality mapping, sterility, and reliability in AR-supported IR practice.

3 Prism System Overview

PRISM is a Vision Pro–based multimodal AR assistant co-designed to support sterile, in-procedure imaging workflows in interventional radiology (IR). Rather than aiming for clinical efficacy claims at this stage, Prism focuses on workflow-centered interaction design: consolidating multi-screen imaging into an in-field AR workspace and enabling hands-free, sterile-first control through voice and hand gestures. Figure 1 summarizes the three core components of the Prism system and their roles within sterile interventional workflows.

Design Goals. Informed by prior reviews of AR systems in IR and multimodal interaction patterns for medical image viewers [15], Prism is guided by four goals: (G1) reducing context switching between the procedural field and external monitors; (G2) preserving sterile workflow by minimizing reliance on non-sterile peripherals and technician-mediated image navigation; (G3) lowering cognitive and physical burden through glanceable, stable layouts; and (G4) supporting key IR tasks during high-attention phases such as needle targeting and insertion.

XR Imaging Workspace and Workflow Context. Prism consolidates imaging content into an AR “cockpit” of anchored virtual displays positioned within the clinician’s field of view (Figure 1a). Depending on the case, displayed content may include DICOM-based scan series, cross-sectional views (e.g., CT/CBCT), and lightweight reference materials. Layouts are configurable across procedure phases, enabling rapid switching between presets while keeping frequently referenced views persistent. In addition to imaging, Prism provides a lightweight procedural context panel (Figure 1c) that surfaces phase-relevant reference information without requiring clinicians to turn away from the sterile field.

Sterile Multimodal Interaction. Interaction in Prism follows an explicit task-to-modality mapping aligned with prior findings in multimodal image navigation [14, 27, 31]. Voice input is used for discrete state changes, such as switching layouts or series, toggling modes, or activating system views, while hand gestures support

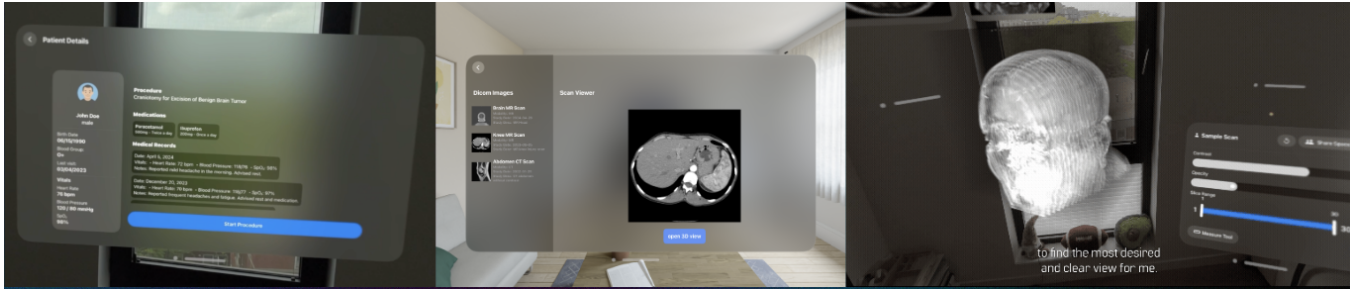


Figure 1: PRISM system overview showing three core components of the AR workspace: (a) a DICOM-based imaging workspace presenting cross-sectional scans within the clinician’s field of view; (b) an optional, on-demand spatial inspection view that allows clinician-controlled exploration of volumetric data without persistent overlay; and (c) a lightweight procedural context panel providing phase-relevant reference information during the intervention. Together, these components illustrate how Prism consolidates imaging and workflow context into a sterile-first, multimodal AR system for interventional radiology.

IR imaging task	Prism modality
Switch layout / series / mode	Voice (discrete)
Activate or dismiss system views	Voice (discrete)
Pan/zoom, navigate slices or time	Hand gestures (continuous)
Reposition, resize, or anchor windows	Hand gestures (continuous)

Table 1: Task–modality mapping in PRISM: voice supports discrete actions, while hand gestures support continuous navigation and manipulation [14, 27, 31].

continuous operations including pan/zoom, slice navigation, and window repositioning (Table 1).

Optional Spatial and Guided Views. For phases requiring deeper inspection or alignment, Prism offers optional, on-demand spatial views (Figure 1b) that allow clinicians to explore volumetric data without introducing persistent overlays. During needle insertion—a high-attention phase—Prism provides a dedicated guided view designed to prioritize stable access to the most relevant imagery while minimizing occlusion of hands, instruments, and anatomy, reflecting known visibility and field-of-view constraints of head-worn AR systems in IR environments [15].

The guided view currently supports a small set of descriptive visual elements, including a persistent reference pane for the active imaging slice, clinician-controlled indicators for viewing angle and depth, and lightweight step or state cues that reflect the current phase of the insertion workflow. These elements are intended to support glanceable orientation rather than automated guidance or accuracy enhancement.

4 Clinician Co-Design Process

PRISM was developed through a clinician-led co-design process with two interventional radiologist (IR). Our goal was to surface workflow constraints in real IR practice and translate them into sterile-first interaction requirements for an AR imaging workspace. The process was informed by principles of contextual inquiry and contextual design, which emphasize understanding work as it is

performed in situ and iteratively interpreting findings with stakeholders [18, 36], as well as participatory and co-design approaches commonly used in clinical HCI to align prototypes with high-stakes workflows [34, 35].

We began with semi-structured workflow interviews (approximately 1–2 hours each) with the two clinician co-authors. Interviews focused on procedural phases and task demands, imaging dependencies across phases, sterile constraints and posture limitations, and coordination patterns with technologists. Clinicians highlighted several high-friction moments, including frequent image series switching during needle targeting, the need to confirm specific cross-sectional views while maintaining instrument position, and reliance on verbal requests to technologists for image navigation during critical phases.

To ground these accounts in situated practice, we conducted contextual observation of an IR procedure workflow from outside the procedure room. Observations focused on the information ecology of the IR suite, including how imaging is distributed across monitors, how clinicians shift attention between the procedural field and displays, and how image switching is performed through control consoles or technologist mediation. We noted repeated attention shifts during targeting and confirmation phases, as well as situations in which sterile posture constrained clinicians’ ability to directly access input devices.

Insights from interviews and observation informed the design and implementation of a Vision Pro prototype, followed by iterative clinician walkthroughs using representative scenarios. During these walkthroughs, clinicians interacted with the XR workspace, voice and gesture controls, and the guided needle insertion view, providing feedback on task–modality mapping, layout stability, occlusion risk, and phase-specific information needs. Across multiple iteration rounds, we refined (1) XR layout orchestration, determining which views should remain persistent versus appear on demand across procedural phases; (2) task–modality mapping, assigning voice input to discrete state changes and hand gestures to continuous navigation [14, 27, 31]; and (3) the guided insertion view, emphasizing glanceable cues and minimizing interference with hands and instruments in line with known AR visibility constraints in IR environments [15].

5 Discussion and Limitations

Prism explores how head-worn AR can be integrated into interventional radiology workflows as an *interaction and workflow design problem*, rather than as a question of visualization novelty or clinical efficacy. While prior work has demonstrated the feasibility of AR systems in IR environments [3, 10, 11, 15], our contribution lies in articulating how sterile constraints, procedural phases, and task structure shape interaction and layout decisions during live procedures.

Workflow-centered AR interaction. Our findings emphasize that the primary opportunity for AR in IR is not persistent augmentation, but the reduction of context switching during high-attention phases. By consolidating imaging and lightweight procedural context into an in-field workspace, Prism foregrounds *when* clinicians need information and *how long* it should remain visible. This perspective shifts AR design away from always-on overlays toward phase-aware, selectively activated views that align with procedural flow.

Sterile-first multimodal task mapping. Prism contributes an explicit task–modality mapping grounded in sterile clinical practice: voice input supports discrete state changes (e.g., layout or series switching), while hand gestures support continuous navigation (e.g., pan/zoom and slice scrubbing). Rather than treating multimodality as an optional interface feature, our design frames it as a workflow tool that reduces reliance on non-sterile peripherals and technician-mediated interaction. This mapping provides a transferable design pattern for multimodal image interaction in other sterile or hands-busy domains.

Glanceable guidance through visual restraint. The guided needle insertion view illustrates a design stance of *visual restraint* in high-stakes AR settings [22, 34]. Instead of continuous or automated overlays, guidance elements are designed to be clinician-controlled, phase-specific, and low-occlusion. This approach reflects tradeoffs among field-of-view limits, occlusion risk, and attention demands that are inherent to head-worn AR in IR environments. Our design highlights that effective in-procedure AR guidance may depend more on selective disclosure and layout stability than on rich graphical augmentation.

Limitations and future work. Prism is an early-stage prototype co-designed with two clinician collaborators, and our insights are qualitative and exploratory. We have not evaluated clinical performance or procedural outcomes. Future work will involve broader clinician feedback, simulation-based usability studies, and systematic assessment of workload and interaction reliability. Nonetheless, Prism demonstrates how interaction-centered, workflow-aware design can inform the development of AR systems for high-stakes clinical environments without overextending claims of efficacy.

6 Conclusion

We presented PRISM, a Vision Pro–based multimodal AR assistant co-designed with interventional radiologists to support sterile, in-procedure imaging workflows. Prism consolidates multi-screen imaging into an in-field XR workspace, enables sterile-first voice and gesture interaction for common imaging operations, and offers a phase-specific guided view for needle insertion that prioritizes glanceability and low occlusion.

This work contributes (1) a workflow-centered AR system prototype design that reconfigures how imaging is accessed and controlled during live procedures, and (2) early design insights into sterile-first multimodal interaction and XR information layout for high-stakes clinical environments. Together, these contributions highlight how careful alignment between workflow demands, interaction modalities, and AR display design can inform the development of practical XR systems for interventional practice.

Acknowledgments

This project was completed under the guidance of Dr. Raul Uppot and Dr. Min Lang (Massachusetts General Hospital, Interventional Radiology). Zeke D’Ascoli, Kida Huang, and Mayank Chadha supported prototype development. Steven Max Patterson and Y. Jenny Wang provided leadership in discussions and event organization.

References

- [1] Abdullah Al-Abcha, Mohamad Alkhouli, Abhiram Prasad, Malcolm Bell, Suraj Yalamuri, Amir Lerman, Rajiv Gulati, and Trevor Simard. 2023. Augmented reality and ultrasound-guided vascular access. *Circulation: Cardiovascular Interventions* 16, 9 (2023). doi:10.1161/CIRCINTERVENTIONS.123.013360
- [2] D Amiras, TJ Hurkxkens, D Figueroa, PJ Pratt, B Pitrola, C Watura, S Rostampour, GJ Shimshon, and M Hamady. 2021. Augmented reality simulator for CT-guided interventions. *European Radiology* 31, 12 (2021), 8897–8902. doi:10.1007/s00330-021-08043-0
- [3] James Baker, Antony Antypas, Prashant Aggarwal, Charlotte Elliott, Robert Baxter, Shwetabh Singh, Naduni Jayasinghe, Daniel Reed, Alexander Boden, Imogen Evans, et al. 2024. Augmented Reality in Interventional Radiology: Transforming Training Paradigms. *Cureus* 16, 2 (Feb. 2024), e54907. doi:10.7759/cureus.54907
- [4] Matthias Barral, Louis Lassalle, François Gardavaud, Raphael Lehrer, Idriss Haffaf, Rémi Agbonon, Olivier Cussenot, and François H. Cornelis. 2024. Virtual Injection Software Reduces Radiation Exposure and Procedural Time of Prostatic Artery Embolization Performed with Cone-Beam CT. *Journal of Vascular and Interventional Radiology* 35, 3 (March 2024), 409–415. doi:10.1016/j.jvir.2023.11.012 Epub 2023-11-25.
- [5] Christopher Bartlett, Noelle LeRoy, Damian Schofield, Jonathan Ford, and Summer Decker. 2020. Assessing the Feasibility of using Augmented Reality to Visualize Interventional Radiology Imagery. In *Proceedings of the 15th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP 2020) - IVAPP*. INSTICC, SciTePress, 169–176. doi:10.5220/0008947701690176
- [6] L. T. O. Bell, R. James, J. A. Rosa, A. Pollentine, G. Pettet, and P. McCoubrie. 2018. Reducing Interruptions during Duty Radiology Shifts, Assessment of Its Benefits and Review of Factors Affecting the Radiology Working Environment. *Clinical Radiology* 73, 8 (Aug. 2018), 759.e19–759.e25. doi:10.1016/j.crad.2018.04.007 Epub 2018-05-28.
- [7] Kristy K. Brock, Stephen R. Chen, Rahul A. Sheth, and Jeffrey H. Siewersden. 2023. Imaging in Interventional Radiology: 2043 and Beyond. *Radiology* 308, 1 (July 2023), e230146. doi:10.1148/radiol.230146
- [8] IV Campbell, Warren A., Jeffrey F. B. Chick, David S. Shin, and Mina S. Makary. 2024. Value of Interventional Radiology and Their Contributions to Modern Medical Systems. *Frontiers in Radiology* 4 (July 2024), 1403761. doi:10.3389/fradi.2024.1403761
- [9] Francois H Cornelis, Leo Razakamanantsoa, Mohamed Ben Ammar, Raphael Lehrer, Idriss Haffaf, Sanaa El-Mouhadi, Francois Gardavaud, Milan Najdawi, and Matthias Barral. 2021. Ergonomics in interventional radiology: Awareness is mandatory. *Medicina* 57, 5 (2021), 500. doi:10.3390/medicina57050500
- [10] Marco Das, Frank Sauer, U Joseph Schoepf, Ali Khamene, Sebastian K Vogt, Stefan Schaller, Ron Kikinis, Eric vanssonenberg, and Stuart G Silverman. 2006. Augmented Reality Visualization for CT-Guided Interventions: System Description, Feasibility, and Initial Evaluation in an Abdominal Phantom. *Radiology* 240, 1 (2006), 230–235. doi:10.1148/radiol.2401040018
- [11] Ahmed Elsakka, Brian J Park, Brett Marinelli, Nathaniel C Swinburne, and Javin Schefflein. 2023. Virtual and augmented reality in interventional radiology: Current applications, challenges, and future directions. *Techniques in vascular and interventional radiology* 26, 3 (2023), 100919.
- [12] Sherise Epstein, Emily H Sparer, Bao N Tran, Qing Z Ruan, Jack T Dennerlein, Dhruv Singhal, and Bernard T Lee. 2018. Prevalence of work-related musculoskeletal disorders among surgeons and interventionalists: a systematic review

- and meta-analysis. *JAMA surgery* 153, 2 (2018), e174947. doi:10.1001/jamasurg.2017.4947
- [13] Gaurav Gadodia, Jeffrey Yanof, Aydan Hanlon, Sara Bustos, Karl West, and Charles Martin III. 2022. Early clinical feasibility evaluation of an augmented reality platform for guidance and navigation during percutaneous tumor ablation. *Journal of Vascular and Interventional Radiology* 33, 3 (2022), 333–338. doi:10.1016/j.jvir.2021.11.014
- [14] N. Gapon, V. Voronin, E. Semenishchev, Y. Ilyukhin, D. Bezuglov, and A. Zeleniskii. 2021. Multimodal Image Inpainting for an Autonomous Robot Navigation Application. In *Multimodal Sensing and Artificial Intelligence: Technologies and Applications II (Proceedings of SPIE, Vol. 11785)*, Ettore Stella (Ed.). SPIE, Online Only, Germany, Article 117850Y. doi:10.1117/12.2594476
- [15] Youssef Ghosn, Javin Schefflein, Yara Jabbour, Debkumar Sarkar, and Francois H. Cornelis. 2025. Augmented reality in interventional radiology: A review of systems, ergonomics, applications and outcomes. *Journal of Medical Extended Reality* 2, 1 (2025), 29941520251377069. doi:10.1177/29941520251377069
- [16] R. F. Grasso, F. Andresiani, C. Altomare, et al. 2021. Lung thermal ablation: Comparison between an augmented reality computed tomography (CT) 3D navigation system (SIRIO) and standard CT-guided technique. *Biology* 10, 7 (2021), 646. doi:10.3390/biology10070646
- [17] Robert T. Held and Thomas T. Hui. 2011. A guide to stereoscopic 3D displays in medicine. *Academic Radiology* 18, 8 (2011), 1035–1048. doi:10.1016/j.acra.2011.04.005
- [18] Karen Holtzblatt and Hugh R. Beyer. 1997. *Contextual Design: Defining Customer-Centered Systems*. Morgan Kaufmann, San Francisco, CA, USA.
- [19] Ferenc A. Jolesz (Ed.). 2014. *Intraoperative Imaging and Image-Guided Therapy*. Springer, New York, NY, USA. doi:10.1007/978-1-4614-7657-3
- [20] George C. Kagadis, Konstantinos Katsanos, Dimitrios Karnabatidis, George Loukos, George C. Nikiforidis, and William R. Hendee. 2012. Emerging Technologies for Image Guidance and Device Navigation in Interventional Radiology. *Medical Physics* 39, 9 (Sept. 2012), 5768–5781. doi:10.1118/1.4747343
- [21] Adrien Kaladji, Aurélien Dumenil, Guillaume Mahé, Miguel Castro, Alain Cardon, Antoine Lucas, and Pascal Haigron. 2015. Safety and accuracy of endovascular aneurysm repair without pre-operative and intra-operative contrast agent. *European Journal of Vascular and Endovascular Surgery* 49, 3 (2015), 255–261. doi:10.1016/j.ejvs.2014.12.003
- [22] Lauren Kowal. 2020. *An Exploration of Anesthesia Providers' Workspace: Impact on Ergonomics and Safety*. Master's thesis. Georgetown University, Washington, DC, USA. <https://repository.digital.georgetown.edu/handle/10822/1059692> Thesis, dated March 26, 2020.
- [23] Ivo Kuhlmann, Markus Kleemann, Philipp Jauer, Achim Schweikard, and Floris Ernst. 2017. Towards X-ray Free Endovascular Interventions – Using HoloLens for On-line Holographic Visualisation. *Healthcare Technology Letters* 4, 5 (2017), 184–187. doi:10.1049/htl.2017.0061
- [24] Marcelo C. F. Macedo and André L. Apolinário. 2023. Occlusion handling in augmented reality: Past, present and future. *IEEE Transactions on Visualization and Computer Graphics* 29, 2 (2023), 1590–1609. doi:10.1109/TVCG.2021.3117866
- [25] Julian Maingard, Hong Kuan Kok, Dinesh Ranatunga, Duncan Mark Brooks, Ronil V Chandra, Michael J Lee, and Hamed Asadi. 2017. The Future of Interventional and Neurointerventional Radiology: Learning Lessons from the Past. *The British Journal of Radiology* 90, 1080 (2017), 20170473. doi:10.1259/bjr.20170473 Epub 2017-10-03.
- [26] André Mewes, Bennet Hensen, Frank Wacker, and Christian Hansen. 2017. Touchless Interaction with Software in Interventional Radiology and Surgery: A Systematic Literature Review. *International Journal of Computer Assisted Radiology and Surgery* 12, 2 (Feb. 2017), 291–305. doi:10.1007/s11548-016-1480-6
- [27] Christopher Nimsy, Daniela Kuhnt, Oliver Ganslandt, and Michael Buchfelder. 2011. Multimodal Navigation Integrated with Imaging. In *Intraoperative Imaging*. Acta Neurochirurgica Supplement, Vol. 109. Springer Vienna, 207–214. doi:10.1007/978-3-211-99651-5_32 First online: 2010-01-01.
- [28] Brian J Park, Stephen J Hunt, Charles Martin III, Gregory J Nadolski, Bradford J Wood, and Terence P Gade. 2020. Augmented and mixed reality: Technologies for enhancing the future of interventional radiology. *Journal of Vascular and Interventional Radiology* 31, 7 (2020), 1074–1082. doi:10.1016/j.jvir.2019.09.020
- [29] Brian J. Park, Sana Shah, Dominik Konik, James Y. Lim, and Timothy Huber. 2023. An Ergonomic Holographic Procedural Monitor for Thyroid Radiofrequency Ablation Using a Mixed-Reality Headset. *Journal of Vascular and Interventional Radiology* 34, 2 (Feb. 2023), 307–310. doi:10.1016/j.jvir.2022.10.027 Epub 2022-10-23.
- [30] Zulfkar Qadrie, Mudassir Maqbool, Mohd Altaf Dar, and Afshana Qadir. 2025. Navigating Challenges and Maximizing Potential: Handling Complications and Constraints in Minimally Invasive Surgery. *Open Health* 6, 1 (2025), 48–51. doi:10.1515/oh-2025-0059
- [31] Antoine Rosset, Luca Spadola, Lance Pyshey, and Osman Ratib. 2006. Navigating the Fifth Dimension: Innovative Interface for Multidimensional Multimodality Image Navigation. *Radiographics* 26, 1 (2006), 299–308. doi:10.1148/rg.261055066
- [32] Luigi A. Solbiati. 2021. Augmented reality: Thrilling future for interventional oncology? *Cardiovascular and Interventional Radiology* 44, 5 (2021), 782–783. doi:10.1007/s00270-021-02801-9
- [33] Justin H Tan, Cherrng Chao, Mazen Zawaideh, Anne C Roberts, and Thomas B Kinney. 2013. Informatics in Radiology: Developing a Touchless User Interface for Intraoperative Image Control during Interventional Radiology Procedures. *Radiographics* 33, 2 (March 2013), E61–E70. doi:10.1148/rg.332125101
- [34] Angelique Taylor, Tauhid Tanjim, Huajie Cao, Jalynn Blu Nicoly, Jonathan I Segal, Jonathan St George, Soyoon Kim, Kevin Ching, Francisco R Ortega, and Hee Rin Lee. 2025. Co-Designing Augmented Reality Tools for High-Stakes Clinical Teamwork. *arXiv preprint arXiv:2502.17295* (Feb. 2025). arXiv:2502.17295 [cs.HC] doi:10.48550/arXiv.2502.17295
- [35] Jasper T. Vermeulen, Stine S. Johansen, Alan G. Burden, and James L. Dwyer. 2025. Removing the Drapes: Reflexive Insights on Research into Real-World Surgical Human-Robot Collaboration. In *Proceedings of the 37th Australian Conference on Human-Computer Interaction (OZCHI 2025)*, Gadigal / Sydney, Australia, 29 November 2025 - 3 December 2025. ACM, 868–875. doi:10.1145/3764687.3769938
- [36] Paraskevas Vezyridis, Stephen Timmons, and Heather Wharrad. 2011. Going Paperless at the Emergency Department: A Socio-Technical Study of an Information System for Patient Tracking. *International Journal of Medical Informatics* 80, 7 (July 2011), 455–465. doi:10.1016/j.ijmedinf.2011.04.001 Epub 2011-05-13.
- [37] Clifford R Weiss and Nima Hafezi-Nejad. 2023. Interventional Radiology: Past, Present, and Future. *Radiology* 308, 1 (July 2023), e230809. doi:10.1148/radiol.230809
- [38] Jian Yang, Jianjun Zhu, Daniel Y Sze, Li Cui, Xiaohui Li, Yanhua Bai, Danni Ai, Jingfan Fan, Hong Song, and Feng Duan. 2020. Feasibility of augmented reality-guided transjugular intrahepatic portosystemic shunt. *Journal of Vascular and Interventional Radiology* 31, 12 (2020), 2098–2103. doi:10.1016/j.jvir.2020.07.025
- [39] Paolo Zaffino, Sara Moccia, Elena De Momi, and Maria Francesca Spadea. 2020. A Review on Advances in Intra-operative Imaging for Surgery and Therapy: Imagining the Operating Room of the Future. *Annals of Biomedical Engineering* 48, 8 (2020), 2171–2191. doi:10.1007/s10439-020-02553-6